Yaksh Dataset Report

Real-World Educational Logic Error Analysis

1. OBJECTIVE

The goal of this phase was to evaluate how multiple large language models classify logical errors on a real-world educational dataset (**Yaksh**), and to understand agreement, disagreement, and `NONE` behavior across closed-source and open-source models using a unified taxonomy (A–F, `NONE`).

2. DATASET AND SETUP

Experimental Setup

- **Dataset:** Yaksh programming submissions.
- **Total cases analyzed:** 81.
- **Input:** Problem description + Student-submitted Python code (often containing mixed solutions/drivers).
- **Models Evaluated (5):**
  - **Closed-Source:** Sonnet, Gemini.
  - **Open-Source:** DeepSeek, Qwen, GPT-OSS.
- **Prompt Constraints:** Strict taxonomy (A–F + `NONE`), Priority order (**D > F > C > A > B > E**), Single-character output.

3. HIGH-LEVEL AGREEMENT RESULTS

Agreement Statistics (N=81)

- **All models agree:** 12 cases
- **Majority agreement ($\geq$3 models):** 52 cases
- **High-conflict (no majority):** 17 cases

Key Observation: Even with a strict taxonomy, $\sim$21% of cases remain high-conflict, indicating *intrinsic ambiguity* rather than noise.

4. PER-MODEL BEHAVIOR SUMMARY

Sonnet & Gemini (Closed)
Strong concentration on **D (Algorithmic)** and **F (Specification)**. Moderate use of `NONE` (11 each). Very limited use of low-level categories (A–C).

DeepSeek (Open)
More diverse spread across A–F. Higher `NONE` usage (19). Less aggressive in forcing a dominant label.

Qwen (Open)
Extremely high `NONE` usage (50 / 81). Very low F usage (6). Acts as a strong *logic-isolation* model.

GPT-OSS (Open)
Similar to closed models on D and F, but with higher `NONE` usage. Acts as a bridge between closed and open reasoning styles.

5. CLOSED VS OPEN MODEL AGREEMENT

- **Closed–Closed (Sonnet vs Gemini):** Agreement on **54 / 81** cases (∼67%). Indicates a strong shared abstraction level.
- **Open–Open:** Strict agreement (all three) in only **15 cases**. Open models are not internally consistent as a group.

6. `NONE` AS A MINORITY SIGNAL

Important Finding: Logic Isolation Strategy

We analyzed cases where a clear majority labeled A–F, but a minority output `NONE`.

- **Qwen dominates this behavior** (23 cases).
- **Interpretation:** `NONE`-as-minority is not random; it reflects a **logic-isolation strategy**, where the model ignores specification violations (F) to focus narrowly on core computation.

7. TWO DISTINCT COLLECTIVE BEHAVIORS

From repeated runs, we observed two stable behaviors on the same input:

Behavior 1: Specification-Dominant Consensus

Result: Most or all models → **F**
Trigger: Extra functions, driver code mixed in, output contract violated.

Behavior 2: Fragmented Interpretation

Result: Outputs split across **F / D /** `NONE`
Cause: Multi-fault submissions (Spec violated + Algo incorrect + Function isolated).
Insight: Reflects different anchoring strategies:

- Specification-first → **F**
- Algorithm-first → **D**
- Function-isolation → **NONE**

8. DEVIATION FROM MAJORITY

- **Qwen:** Deviates most (ignores F and D).
- **DeepSeek:** Deviates between D and F.
- **GPT-OSS:** Deviates mainly via `NONE` or F.
- **Sonnet/Gemini:** Deviate rarely.

This reinforces that disagreement is systematic and model-dependent.

9. CORE CONCLUSIONS FROM YAKSH DATA

General Rules

1. Logical error classification on real student code is **not always single-label stable**.
2. **Closed models** emphasize specification fidelity; **Open models** vary widely (some isolate logic).
3. `NONE` is a meaningful signal, not a failure case.
4. Multi-fault submissions expose the limits of forced single-label classification.

COMBINED MODEL AGREEMENT HEATMAP

Visualizing pairwise agreement percentages across 68 logical error cases. Darker cells indicate higher consensus.

Model	Claude	Gemini	DeepSeek	Qwen	GPT-OSS
Claude	-	66.7	53.1	38.3	63.0
Gemini	66.7	-	54.3	33.3	63.0
DeepSeek	53.1	54.3	-	33.3	43.2
Qwen	38.3	33.3	33.3	-	34.6
GPT-OSS	63.0	63.0	43.2	34.6	-

Table 1: ***Pairwise Agreement percentages (darker = higher agreement). Note the strong cluster between Claude, Gemini, and GPT-OSS.***

Insight from Heatmap

- **Strongest Pair:** Claude ↔ Gemini (66.7%)
- **Weakest Pair:** Qwen ↔ Gemini (33.3%)
- **Cluster:** Claude, Gemini, and GPT-OSS form a relatively cohesive group ($>$ 60% agreement), while DeepSeek and Qwen act as outliers.

10. FINAL TAKEAWAY

Practical Implications

The Yaksh dataset reveals that logical error classification is inherently ambiguous for multi-fault student submissions. Model disagreement reflects fundamentally different reasoning anchors—**specification, algorithm, or isolation**—rather than annotation noise or model instability.

10 codes running again and again

1 ITERATION DISAGREEMENT ANALYSIS (STABILITY CHECK)

Analyzing whether models change their minds when allowed to list multiple errors (Iter-2) or when asked to re-evaluate (Iter-3).

1. Disagreement: Iteration 1 → Iteration 2 (Blind Guesses)

Definition: A disagreement occurs when the single label chosen in Iter-1 is **NOT** contained inside the label set of Iter-2. This indicates a "blind guess" where the initial intuition contradicts the deeper analysis.

Blind Guess Count (Lower is Better)

- **Sonnet:** 1
- **Gemini: 0** (Perfect Stability)
- **DeepSeek: 3** (Highest Instability)
- **Qwen:** 1
- **GPT-OSS:** 1

Insight: Anchoring Stability

- **DeepSeek** is the most likely to change its decision after deeper analysis.
- **Gemini** never blindly guessed—its Iter-1 choice was always supported in Iter-2.
- **Conclusion:** Disagreement between Iter-1 and Iter-2 is rare overall but concentrated in open models.

2. Disagreement: Iteration 2 → Iteration 3 (Intra-Model Instability)

Definition: Captures whether the model remains consistent when asked to list errors again. This measures *intra-model instability*.

Unstable Cases per Model (Iter-2 $\neq$ Iter-3)

(Dataset size = 10 codes)

- **DeepSeek: 6** (Most Unstable)
- **GPT-OSS:** 3
- **Sonnet:** 3
- **Qwen:** 2
- **Gemini: 0** (Perfectly Stable)

Insight: Structural Ambiguity

- Iter-2 → Iter-3 disagreement is **significantly more common** than Iter-1 → Iter-2 disagreement.
- **DeepSeek** fluctuates the most when listing multiple errors.
- **Gemini** demonstrates robust internal consistency.

3. Model Instability Ranking

Ranking models from most unstable to most stable based on cumulative behavior.

1. **DeepSeek (Most Unstable):** Most blind guesses; most Iter-2 $\leftrightarrow$ Iter-3 changes. Strongly anchor-sensitive.
2. **GPT-OSS:** Moderate instability; oscillates between F and multi-label sets.
3. **Sonnet:** Rare blind guesses but some expansion instability.
4. **Qwen:** Few switches, but generates large label sets when it does switch.
5. **Gemini (Most Stable):** No blind guesses. No Iter-2 $\leftrightarrow$ Iter-3 instability.

2 ITERATION-WISE MODEL AGREEMENT HEATMAPS

Comparing pairwise agreement across three iterations. Agreement increases significantly when models are allowed to list multiple errors (Iter-2/3).

Iteration 1: Single-Label Constraint (Strict)

Low agreement overall. DeepSeek and GPT-OSS struggle to align with others.

Model	Sonnet	Gemini	DeepSeek	Qwen	GPT-OSS
Sonnet	-	60.0	50.0	50.0	70.0
Gemini	60.0	-	50.0	70.0	50.0
DeepSeek	50.0	50.0	-	40.0	30.0
Qwen	50.0	70.0	40.0	-	60.0
GPT-OSS	70.0	50.0	30.0	60.0	-

Iteration 2: Multi-Label (Set Overlap)

Agreement jumps significantly. Qwen and GPT-OSS reach 100% overlap with Sonnet, indicating their single-label choice was valid but partial.

Model	Sonnet	Gemini	DeepSeek	Qwen	GPT-OSS
Sonnet	-	80.0	70.0	**100.0**	**100.0**
Gemini	80.0	-	40.0	**100.0**	80.0
DeepSeek	70.0	40.0	-	80.0	70.0
Qwen	**100.0**	**100.0**	80.0	-	**100.0**
GPT-OSS	**100.0**	80.0	70.0	**100.0**	-

Iteration 3: Multi-Label (Re-Evaluation)

Near-perfect consensus across most models, confirming that disagreement is largely an artifact of forced single-label selection.

Model	Sonnet	Gemini	DeepSeek	Qwen	GPT-OSS
Sonnet	-	80.0	90.0	**100.0**	**100.0**
Gemini	80.0	-	80.0	**100.0**	90.0
DeepSeek	90.0	80.0	-	90.0	90.0
Qwen	**100.0**	**100.0**	90.0	-	**100.0**
GPT-OSS	**100.0**	90.0	90.0	**100.0**	-

Key Insight: The "Artificial Conflict" of Single Labels

- **Iter-1 (Single-Label):** DeepSeek and GPT-OSS show very low agreement (30%) with closed models.
- **Iter-2 (Multi-Label):** Agreement skyrockets. **Qwen and GPT-OSS reach 100% agreement** with Sonnet.
- **Conclusion:** Models often see the *same* errors but prioritize them differently. When allowed to list all errors (Iter-2/3), the apparent disagreement vanishes.

Summary & Takeaway

Meeting-Ready Summary

"Disagreement between Iteration-1 and Iteration-2 is rare and mostly limited to open models, with **DeepSeek** showing the highest blind-guess rate. In contrast, disagreement between Iteration-2 and Iteration-3 is much more common, indicating intra-model instability when identifying multiple errors. DeepSeek is the most unstable overall, while **Gemini** is fully stable across all iterations. This shows that instability arises not from randomness, but from multi-anchor ambiguity in the code."

Core Takeaway

Models rarely guess blindly, but many struggle to stabilize once multiple valid error interpretations are allowed.

Comprehensive Human–Model Evaluation

Across Three Logical Error Taxonomies

EXPERIMENTAL SETUP

A total of **100 buggy Python programs** were evaluated across three independently designed logical error taxonomies. The evaluation involved six state-of-the-art models: **Claude Sonnet 4.5, Gemini 2.5 Flash, GPT-5.2, DeepSeek v3.2, Qwen3-Coder,** and **GPT-OSS 120B**.

Harshit Taxonomy Global correctness oriented.

Tanishi Taxonomy Structural-local focus.

Yuvraj Taxonomy Pedagogically granular.

Methodology:

Iteration 1: Single dominant label generation.

Iteration 2: Multi-label output generation.

Baseline: Human labels were manually assigned for all three taxonomies.

3 1. SINGLE → MULTI CONSISTENCY

This metric measures how often the single-label prediction was contained inside the multi-label prediction, serving as a proxy for internal model stability.

Table 2: Single-to-Multi Consistency Rate

Model	Har	Tan	Yuv
Sonnet	0.91	0.55	0.94
Gemini	0.89	0.39	0.93
GPT-5.2	0.92	0.75	0.99
DeepSeek	0.88	0.50	0.78
Qwen	0.93	0.52	0.95
GPT-OSS	0.92	0.66	0.90

Insight

- **Stability:** Highest stability observed in the Yuvraj taxonomy.
- **Instability:** Tanishi taxonomy causes instability (significantly more label expansion).
- **Winner:** GPT-5.2 shows the strongest internal consistency overall.

4 2. MULTI-LABEL VS HUMAN (EXACT MATCH)

This measures the strict alignment between model predictions and human ground truth.

Table 3: Exact Match Rate (Multi vs Human)

Model	Har	Tan	Yuv
Sonnet	0.20	0.11	0.13
Gemini	0.21	0.03	0.09
GPT-5.2	0.19	0.12	0.17
DeepSeek	0.19	0.00	0.05
Qwen	0.21	0.00	0.12
GPT-OSS	0.21	0.11	0.10

Insight

- All models struggle to achieve high exact match scores.
- **Best Fit:** Har taxonomy yields the highest alignment with humans.
- **Worst Fit:** Tanishi taxonomy produces very low exact match due to structural granularity.

Table 4: Multi-label vs Human Agreement under Harshit Taxonomy

Model	Exact (%)	Partial (%)	No Match (%)
sonnet	**36.0**	47.0	17.0
gemini	34.0	45.0	21.0
deepseek	32.0	48.0	20.0
gpt_oss	31.0	53.0	16.0
chat_gpt	30.0	54.0	16.0
qwen	30.0	**57.0**	**13.0**

Table 5: Multi-label vs Human Agreement under Tanishi Taxonomy

Model	Exact (%)	Partial (%)	No Match (%)
chat_gpt	**12.0**	**56.0**	32.0
sonnet	11.0	55.0	34.0
gpt_oss	11.0	50.0	39.0
gemini	3.0	43.0	54.0
deepseek	0.0	48.0	52.0
qwen	0.0	50.0	50.0

Table 6: Multi-label vs Human Agreement under Yuvraj Taxonomy

Model	Exact (%)	Partial (%)	No Match (%)
chat_gpt	**20.0**	66.0	**14.0**
gpt_oss	13.0	**68.0**	19.0
qwen	15.0	64.0	21.0
sonnet	15.0	64.0	21.0
gemini	10.0	66.0	24.0
deepseek	8.0	67.0	25.0

5 3. SINGLE-LABEL CONTAINMENT IN HUMAN LABELS

This analyzes whether the single dominant label predicted by the model is present within the set of human-assigned labels.

Table 7: Single Label Overlap with Human

Model	Har	Tan	Yuv
Sonnet	0.40	0.64	0.39
Gemini	0.40	0.56	0.41
GPT-5.2	0.40	0.61	0.40
DeepSeek	0.38	0.39	0.38
Qwen	0.40	0.55	0.43
GPT-OSS	0.42	0.51	0.38

Insight

- **Tanishi Taxonomy:** Shows higher containment, indicating models successfully catch at least one structural flaw.
- **Har & Yuv Taxonomies:** Show lower containment due to conceptual dominance and abstraction.

6 4. PAIRWISE MODEL AGREEMENT (MULTI-LABEL)

6.1 Har Taxonomy

- Highest agreement pair: **GPT-5.2 ↔ GPT-OSS = 0.83**
- Closed models agreement average: ≈ 0.69
- Open model agreement average: ≈ 0.70

6.2 Tan Taxonomy

Agreement is significantly lower across the board.

- Average cross-model agreement: ≈ 0.49
- *Note: DeepSeek diverges most in this taxonomy.*

6.3 Yuv Taxonomy

Moderate agreement observed.

- Average: ≈ 0.58
- *Note: GPT-5.2 and GPT-OSS remain the most aligned pair.*

7 5. CROSS-TAXONOMY OBSERVATIONS

7.1 Taxonomy Behavior Differences

Har Taxonomy Encourages global reasoning. Produces higher inter-model agreement.

Tanishi Taxonomy Encourages structural reasoning. Produces high disagreement and label expansion.

Yuvraj Taxonomy Pedagogically granular. Models expand labels but remain internally consistent.

7.2 Model Stability Ranking (Overall)

GPT-5.2 > GPT-OSS > Sonnet > Gemini > Qwen > DeepSeek

7.3 Key Findings

- Internal consistency (Single → Multi) does **not** imply human correctness.
- Exact match rates remain low across all taxonomies.
- Structural taxonomies amplify disagreement among models.
- Closed models exhibit slightly higher mutual agreement.

8 6. FINAL INTERPRETATION

Insight

[title=Conclusion]

1. Models are stable but not deeply aligned with human reasoning.
2. **Har taxonomy** best reflects collective model behavior.
3. **Tanishi taxonomy** best exposes structural disagreement.
4. **Yuvraj taxonomy** reveals conceptual expansion tendencies.

Overall, model agreement improves when taxonomy is coarse-grained and declines when taxonomy is structurally granular.

9 7. CLOSED VS OPEN MODEL COMPARATIVE ANALYSIS

We categorize the evaluated models as follows:

Closed-source models:

- Claude Sonnet 4.5
- Gemini 2.5 Flash
- GPT-5.2

Open-weight models:

- DeepSeek v3.2
- Qwen3-Coder
- GPT-OSS 120B

9.1 7.1 Exact Match with Human (Multi-label)

Table 8: Average Exact Match Rate (Multi vs Human)

Group	Har	Tan	Yuv
Closed Models Avg	0.20	0.09	0.13
Open Models Avg	0.20	0.04	0.09

Observation:

- Closed models slightly outperform open models in structural (Tan) and pedagogical (Yuv) taxonomies.
- Performance gap is small in Har taxonomy.

9.2 7.2 Internal Consistency (Single → Multi)

Table 9: Average Internal Consistency

Group	Har	Tan	Yuv
Closed Avg	0.91	0.56	0.95
Open Avg	0.91	0.56	0.88

Observation:

- Internal stability is similar across groups.
- Open models show slightly lower stability under Yuv taxonomy.

10 8. PAIRWISE AGREEMENT HEATMAPS

Pairwise agreement is measured using average Jaccard similarity across 100 questions.

10.1 8.1 Har Taxonomy (Multi-label)

Figure 1: Pairwise Model Agreement – Har Taxonomy (Multi)

Observation:

- Highest agreement observed between GPT-5.2 and GPT-OSS.
- Closed models cluster closely.
- Overall agreement relatively high ($\approx$ 0.69 average).

10.2 8.2 Tanishi Taxonomy (Multi-label)

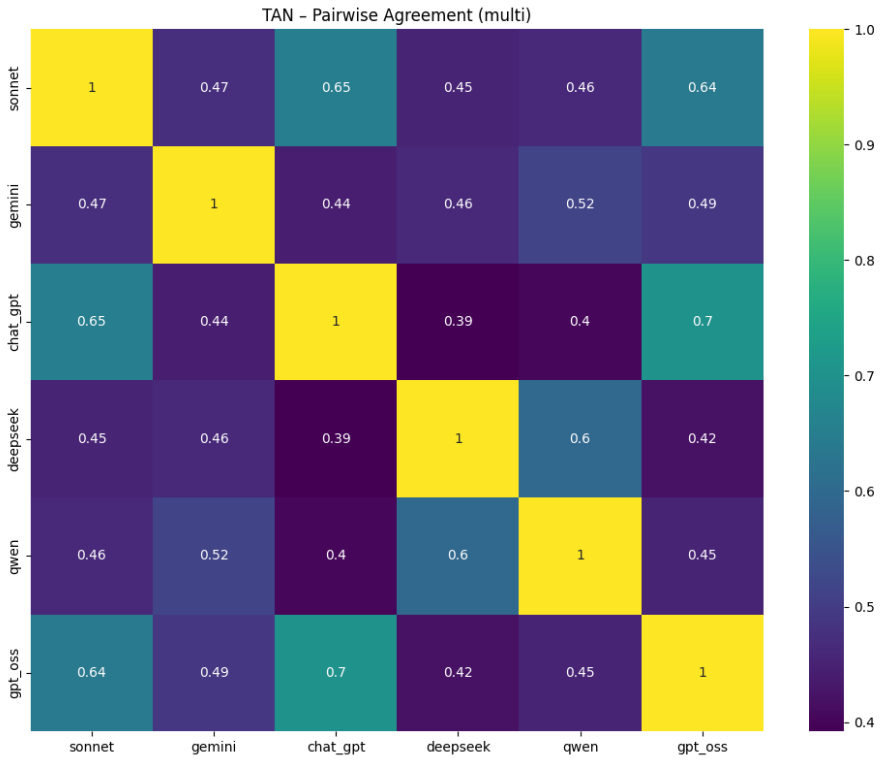

Figure 2: Pairwise Model Agreement – Tanishi Taxonomy (Multi)

Observation:

- Significant drop in cross-model agreement.
- Structural taxonomy increases disagreement.
- DeepSeek shows highest divergence.

10.3 8.3 Yuvraj Taxonomy (Multi-label)

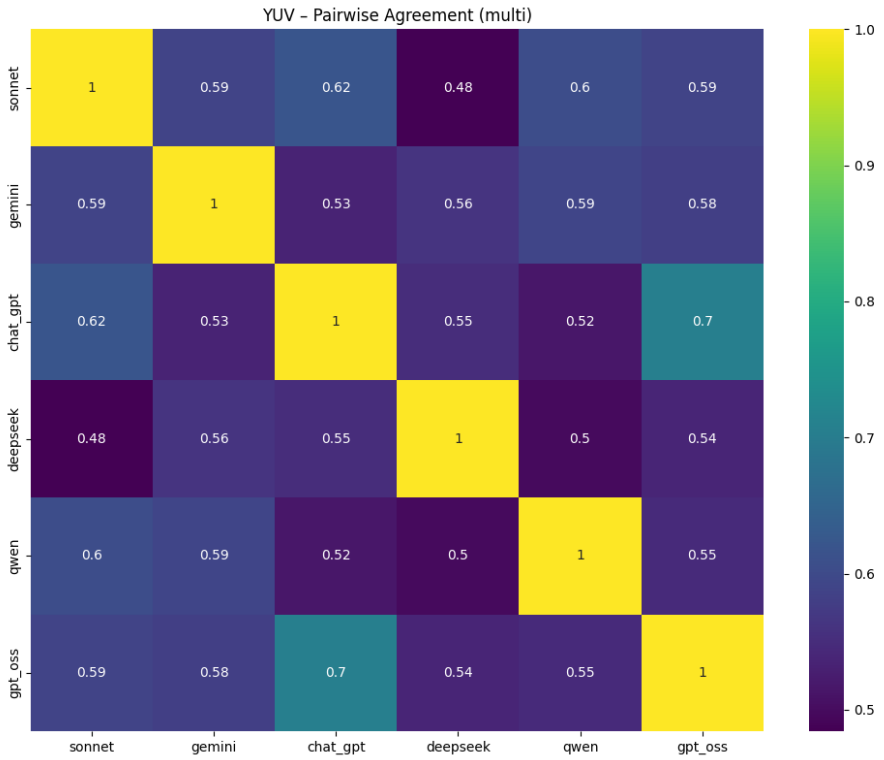

Figure 3: Pairwise Model Agreement – Yuvraj Taxonomy (Multi)

Observation:

- Moderate agreement.
- GPT-5.2 and GPT-OSS remain consistently aligned.
- Open and closed models intermix without strong clustering.

11 9. CROSS-TAXONOMY COMPARATIVE INSIGHT

11.1 Agreement Trend

$$\textbf{Har} > \textbf{Yuv} > \textbf{Tan}$$

This indicates that:

- Coarse-grained conceptual taxonomies yield higher agreement.
- Structural-granular taxonomies amplify disagreement.
- Multi-label expansion increases divergence under fine-grained classification.

11.2 Model Behaviour Summary

- GPT-5.2 and GPT-OSS consistently show strongest pairwise agreement.
- DeepSeek shows highest volatility under structural taxonomy.
- All models demonstrate stability internally but moderate alignment with human judgment.

12 MULTI-LABEL MODEL EVALUATION AGAINST HUMAN ANNOTATIONS

This section evaluates model predictions under a multi-label setting using four progressively relaxed agreement criteria against human annotations (HAR, TAN, YUV).

12.1 Evaluation Criteria

Let P denote the predicted label set and M the manually annotated label set.

- **Case A (Prediction $\subseteq$ Manual):** All predicted labels are correct (no extraneous labels).
- **Case B (Manual $\subseteq$ Prediction):** All human labels are covered by the model (may include extras).
- **Case C (Overlap > 0):** At least one correct label is predicted.
- **Case D (Jaccard Score):**

$$J(P, M) = \frac{|P \cap M|}{|P \cup M|}$$

12.2 HAR Multi vs Human

Model	Case A (%)	Case B (%)	Case C (%)	Mean Jaccard
ChatGPT	21	41	48	0.3185
DeepSeek	23	37	45	0.2972
Gemini	25	39	45	0.3325
GPT-OSS	24	41	49	0.3270
Qwen	24	**46**	**52**	**0.3417**
Sonnet	24	40	50	0.3333

Table 10: Multi-label vs Human Agreement (HAR)

Observations (HAR)

- Exact containment (Case A) is relatively low across all models (21–25%).
- Qwen achieves the highest relaxed overlap (Case C = 52%) and highest Jaccard score (0.3417).
- Case B values are consistently higher than Case A, indicating models tend to over-predict rather than under-predict.

12.3 TAN Multi vs Human

Observations (TAN)

- ChatGPT achieves the highest overlap rate (68%) and highest Jaccard (0.396).
- DeepSeek and Qwen show significantly weaker alignment under TAN's taxonomy.
- Large gap between Case C and Case A suggests frequent partial correctness but poor strict containment.

Model	Case A (%)	Case B (%)	Case C (%)	Mean Jaccard
ChatGPT	19	43	**68**	**0.3960**
DeepSeek	5	25	48	0.1760
Gemini	18	13	46	0.2132
GPT-OSS	22	30	61	0.3438
Qwen	3	25	50	0.1777
Sonnet	19	41	66	0.3602

Table 11: Multi-label vs Human Agreement (TAN)

12.4 YUV Multi vs Human

Model	Case A (%)	Case B (%)	Case C (%)	Mean Jaccard
ChatGPT	20	39	**56**	**0.3278**
DeepSeek	20	24	50	0.2384
Gemini	20	23	48	0.2608
GPT-OSS	13	37	51	0.2810
Qwen	19	29	51	0.2633
Sonnet	19	24	47	0.2757

Table 12: Multi-label vs Human Agreement (YUV)

Observations (YUV)

- ChatGPT demonstrates strongest performance under YUV taxonomy.
- Case A values remain consistently low (13–20%), reinforcing difficulty of strict multi-label containment.
- Partial overlap dominates across all models.

12.5 Cross-Taxonomy Insights

- Strict containment (Case A) rarely exceeds 25% across taxonomies.
- Overlap-based agreement (Case C) consistently ranges between 45–68%.
- TAN taxonomy appears more aligned with ChatGPT and Sonnet.
- HAR taxonomy produces more uniform performance across models.
- YUV taxonomy yields moderate but stable agreement levels.

13 COMPARATIVE MULTI-LABEL AGREEMENT ANALYSIS

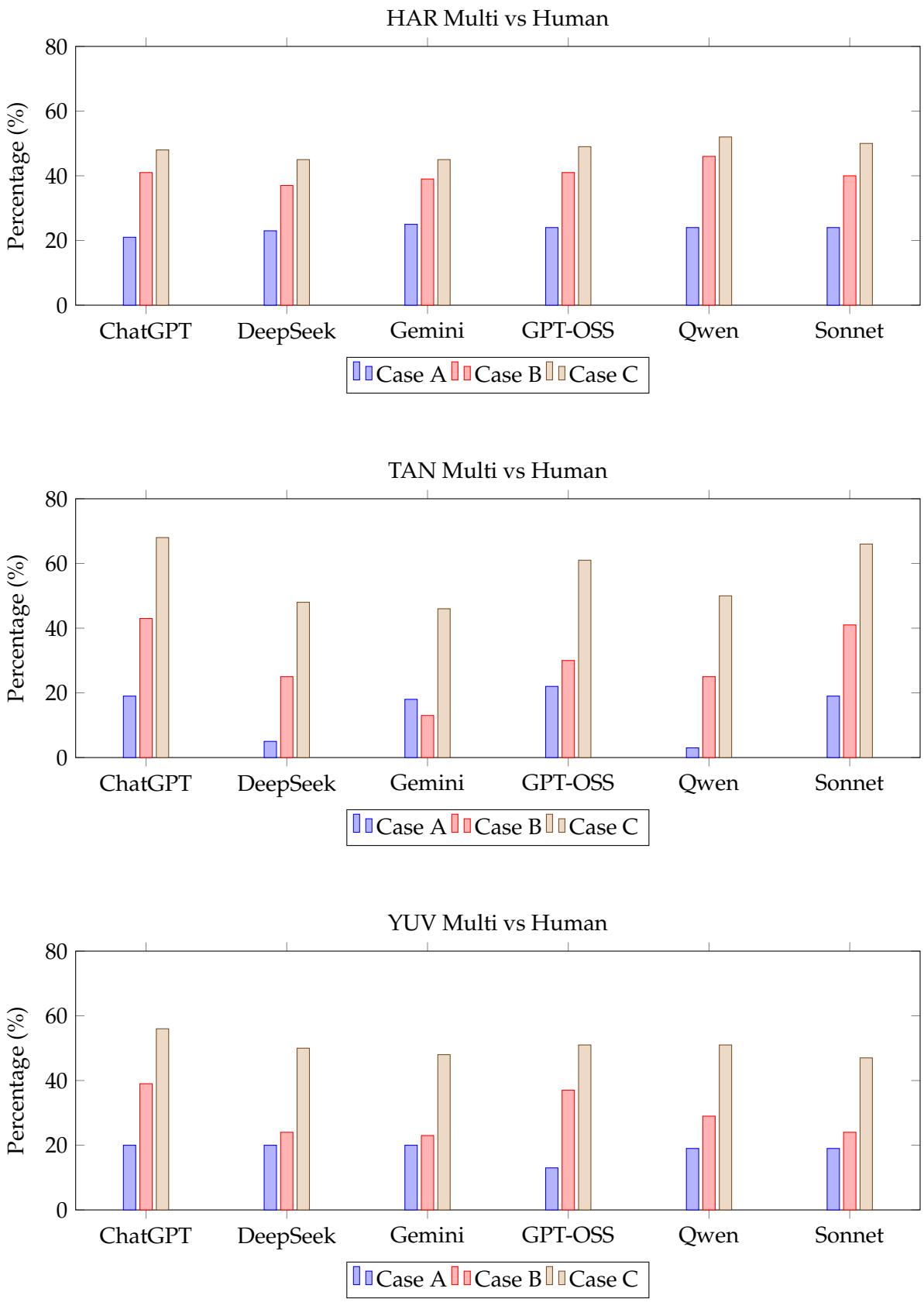

Figure 4: Grouped comparison of multi-label agreement cases across HAR, TAN, and YUV taxonomies.

13.1 Key Interpretation

The evaluation demonstrates that while exact multi-label correctness remains limited, models frequently capture at least part of the human-identified error taxonomy. Performance variation across HAR, TAN, and YUV suggests that taxonomy structure and labeling consistency significantly influence measured agreement.

Overall, models demonstrate stronger behavior under relaxed overlap conditions than strict subset containment, indicating partial but not comprehensive error understanding.

14 RANKING STABILITY ACROSS TAXONOMIES

To evaluate robustness across error categorisation schemes, a weighted composite score was computed for each model using partial-overlap–aware metrics (Cases A–C). Models were then ranked independently for each taxonomy (HAR, TAN, YUV).

14.1 Model Rankings Across Taxonomies

Model	HAR	TAN	YUV
chat_gpt	4	1	1
deepseek	6	5	4
gemini	5	6	5
gpt_oss	3	3	2
qwen	1	4	3
sonnet	2	2	6

Table 13: Model ranking across taxonomies based on weighted composite agreement score (lower rank indicates better performance).

Observations. Performance varies substantially across taxonomies. Qwen performs best under HAR but drops under TAN and YUV. ChatGPT performs strongest under TAN and YUV but not HAR. Sonnet performs consistently strong under HAR and TAN but drops sharply in YUV.

This indicates taxonomy sensitivity: model performance is not invariant to categorisation structure.

14.2 Rank Stability Analysis

To quantify robustness, we compute the standard deviation of rank positions across taxonomies.

Model	Rank Std. Dev
chat_gpt	1.7321
deepseek	1.0000
gemini	0.5774
gpt_oss	0.5774
qwen	1.5275
sonnet	2.3094

Table 14: Ranking stability across taxonomies. Lower standard deviation indicates more consistent performance.

Interpretation.

- **Most Stable Models:** Gemini and GPT-OSS (std = 0.5774). They exhibit consistent relative performance across all categorisation schemes.
- **Moderately Stable:** DeepSeek (std = 1.0).

- **High Variability:** ChatGPT and Qwen show taxonomy-dependent performance shifts.
- **Least Stable:** Sonnet (std = 2.3094), indicating strong sensitivity to taxonomy design.

14.3 Key Findings

1. Model ranking is highly dependent on the error taxonomy used.
2. No single model dominates across all categorisation schemes.
3. Stability analysis reveals that consistency is distinct from peak performance.
4. Taxonomy structure influences evaluation outcomes significantly.

14.4 Implications

These findings demonstrate that:

- Model evaluation in multi-label error detection is taxonomy-sensitive.
- Performance claims must specify the categorisation scheme.
- Stable models may be preferable in educational deployment scenarios.
- High variability suggests interaction between model reasoning patterns and taxonomy granularity.

15 ANOMALY DETECTION AND MANUAL VERIFICATION ANALYSIS

This section presents anomaly-based diagnostics to identify structural weaknesses in model predictions and potential inconsistencies in human annotations across the three taxonomies: HAR, TAN, and YUV.

15.1 Full Disagreement Cases

Full disagreement refers to questions where no consistent majority pattern exists among model predictions.

Taxonomy	Number of Full Disagreement Questions
HAR	15
TAN	51
YUV	16

Table 15: Full disagreement cases across taxonomies

Observation: The TAN taxonomy produces substantially higher disagreement. This indicates that its structural granularity increases fragmentation among models. HAR and YUV demonstrate relatively higher inter-model consistency.

15.2 Model Predictions Outside Human Labels

This anomaly captures cases where a model predicts at least one label not present in the corresponding human annotation.

- **HAR:** Frequent mismatches involve categories F (Specification) and D (Algorithmic Strategy).
- **TAN:** Mismatches often involve structural labels such as E, G, and A.
- **YUV:** Wide dispersion across labels including J, I, H, and D.

Interpretation: HAR exhibits systematic over-attribution of specification errors by models. TAN shows structural misalignment. YUV reveals conceptual divergence and semantic ambiguity.

15.3 High Cardinality Human Annotations

High-cardinality cases are those where the human annotation contains multiple labels.

Implication: Multi-label density correlates strongly with model disagreement. These cases represent complex or ambiguous code samples and serve as stress tests for taxonomy robustness.

15.4 High Difficulty Questions

High difficulty questions are those where a large number of distinct labels appear across model predictions.

Interpretation: These samples demonstrate high entropy and serve as diagnostic benchmarks for:

- Taxonomy sensitivity

- Model calibration under ambiguity
- Inter-model consistency stress testing

15.5 Strong Consensus Against Human Annotation

Strong consensus against human annotation refers to cases where a majority of models agree with each other but disagree with the manual label.

Significance: These cases are critical for:

- Detecting potential human annotation errors
- Identifying systematic bias in taxonomy application
- Understanding collective model reasoning behavior

Such cases warrant manual re-evaluation and may indicate either:

1. Collective model correction of human oversight, or
2. Shared model bias.

15.6 Summary Insight

- TAN taxonomy maximizes structural disagreement.
- HAR taxonomy produces higher stability and inter-model alignment.
- YUV taxonomy increases conceptual dispersion.
- Multi-label density significantly increases prediction entropy.
- Several cases suggest potential human annotation inconsistencies.

Overall, anomaly detection confirms that taxonomy design directly influences both model agreement and human-model alignment, highlighting the importance of evaluation framework selection in logical error classification studies.

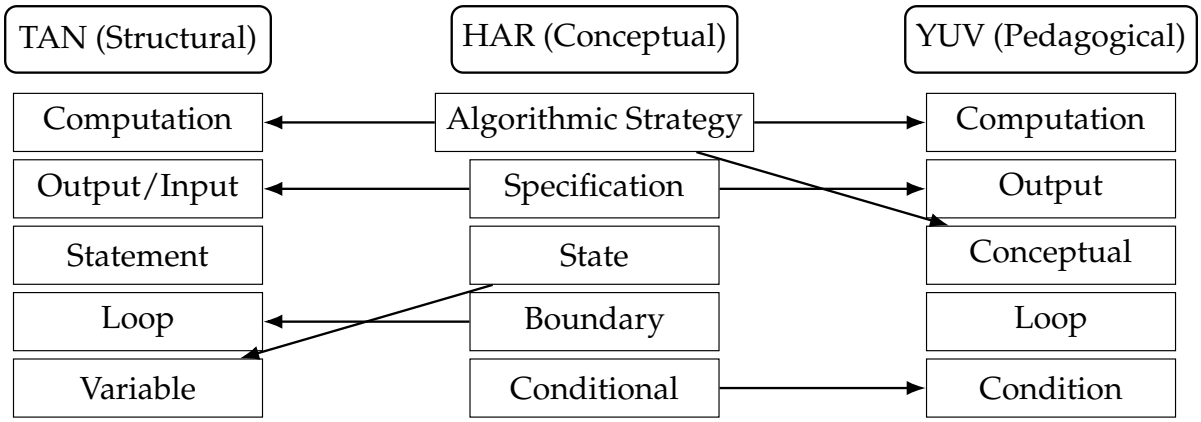

Figure 5: Compact structural alignment between HAR (conceptual), TAN (structural), and YUV (pedagogical) taxonomies.